COMPARATIVE CONSTRUCTION AND ANALYSIS OF NCLDV PROTEIN SIMILARITY NETWORKS USING BLASTP, MMSEQS2 AND SWIPE

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Large DNA viruses have complex and variable gene repertoires, making network-based approaches useful for studying protein relatedness beyond single-marker phylogenies. In this project, proteins from the viral phylum *Nucleocytoviricota*, historically referred to as nucleo-cytoplasmic large DNA viruses (NCLDV), were compared using all-vs-all BLASTP, MMseqs2 and SWIPE searches. The resulting alignments were standardized, filtered under multiple E-value thresholds and converted into undirected protein similarity networks. Duplicate hits between the same protein pair were collapsed primarily by highest bit score, while E-value, percent identity and alignment length were retained for filtering and quality assessment.

The three tools differed strongly in the number of recovered similarity edges, especially in the raw outputs. Stricter E-value filtering reduced these differences and increased cross-tool agreement. At the primary threshold of E-value <= 1e-5, most integrated edges were supported by at least two tools, and the largest support category consisted of edges detected by all three methods. Consensus filtering increased alignment-quality metrics but fragmented the network into smaller high-confidence similarity clusters.

Overall, the results show that NCLDV protein similarity network construction is sensitive to tool choice, threshold selection and edge-selection strategy. A multi-parameter approach combining E-value filtering, bit-score-based edge selection and cross-tool consensus provides a more robust basis for comparing large viral protein similarity networks.

Introduction

Large and giant DNA viruses provide an important system for comparative genomics because their genomes contain both conserved viral core genes and more variable, lineage-specific genes. Nucleo-cytoplasmic large DNA viruses (NCLDV), also referred to in more recent literature as part of the *Nucleocytoviricota*, include several families of large eukaryotic DNA viruses that encode proteins involved in DNA replication, transcription, genome processing and virion morphogenesis[1]. Comparative analyses of NCLDV genomes have shown that, although only a small number of genes are conserved across all members, many additional genes are shared across multiple viral families and can be grouped into Nucleo-Cytoplasmic Virus Orthologous Groups (NCVOGs)[1].

The evolutionary analysis of large DNA viruses is challenging because viral genomes are shaped not only by vertical inheritance, but also by extensive gene gain, gene loss and gene exchange. For this reason, traditional phylogenetic approaches based on a small number of universal marker genes are often insufficient for broad-scale viral evolution. Network-based approaches provide a complementary framework because they can represent shared genes or sequence similarity relationships across many genomes without requiring that all viruses contain the same marker genes[2].

Previous work on dsDNA viruses has shown that gene-sharing networks can reveal modular and hierarchical structure in the virosphere. In such networks, densely connected modules can correspond to groups of related viruses or shared gene repertoires, while highly connected genes can act as hubs linking otherwise distinct modules[2]. This is consistent with general concepts from network biology, where nodes and edges are used to represent biological entities and their relationships, and where properties such as degree, connected components and modularity can be used to describe large biological systems[3].

In the present analysis, the biological question was approached at the protein similarity level. Instead of directly constructing a genome-gene bipartite network, all-vs-all protein sequence comparisons were used to construct protein similarity networks from *Nucleocytoviricota*/NCLDV protein sequences. In these networks, nodes represent proteins and edges represent detected sequence similarity relationships between protein pairs. This type of representation can highlight groups of related proteins, densely connected similarity clusters and high-degree proteins with many sequence similarity relationships.

A central aim of the analysis was to compare how different sequence search methods affect the resulting protein similarity network. BLASTP, MMseqs2 and SWIPE were used as three distinct alignment/search tools. Because these tools differ in their search strategies, speed and reporting behavior, the analysis did not rely on one raw output alone. Instead, multiple alignment parameters were considered, including E-value, bit score, percent identity and alignment length. Several E-value thresholds were tested, and duplicate alignments between the same protein pair were collapsed primarily by highest bit score, while lowest E-value was retained as a sensitivity comparison.

This multi-parameter strategy was used to obtain a more robust comparison between the tools. E-value filtering was used to control the stringency of detected similarity relationships, bit score was used to prioritize the strongest representative alignment between protein pairs, and percent identity together with alignment length was retained to characterize alignment quality. In addition, the overlap between BLASTP, MMseqs2 and SWIPE was used as an independent support criterion: edges detected by multiple tools were interpreted as higher-confidence similarity relationships than tool-specific edges.

The aim of this project was therefore to construct and compare NCLDV protein similarity networks generated from three all-vs-all sequence comparison methods, to evaluate how threshold choice and tool support influence network structure, and to identify high-confidence consensus similarity relationships. By combining sequence similarity search, parameter sensitivity analysis and network-level summaries, the project connects the biological problem of viral protein relatedness with graph-based approaches used in network biology.

Methods

Input data and computational environment

The analysis was based on a protein FASTA file containing *Nucleocytoviricota*/NCLDV protein sequences. The initial all-vs-all sequence comparison workflow followed the course practical provided for the sequence alignment assignment, in which *Nucleocytoviricota* protein sequences were compared with BLASTP, SWIPE and MMseqs2 on the LiSC Life Science Compute Cluster. The cluster environment was used for the computationally intensive alignment step, while downstream filtering, network construction, quality control and visualization were performed after transferring the output files to the local project directory.

Sequence comparison tools

Three sequence comparison tools were used: BLASTP, MMseqs2 and SWIPE. BLASTP was included as a widely used heuristic protein similarity search method[4]. MMseqs2 was included because it is designed for very fast large-scale sequence searching and clustering[5]. SWIPE was included as a fast implementation of Smith–Waterman-based database searching[6]. Comparing all three tools allowed the analysis to evaluate how tool choice influences the resulting protein similarity network.

Alignment output standardization and quality control

The raw alignment outputs were converted into a common tabular format corresponding to the BLAST m8/outfmt6-like structure. The retained columns included query identifier, subject identifier, percent identity, alignment length, E-value and bit score. These parameters were selected because they capture different aspects of sequence similarity: percent identity describes residue identity within the alignment, E-value describes statistical significance, and bit score provides a normalized alignment score[8].

Self-hits were removed before network construction. A self-hit was defined as an alignment where the query and subject identifiers referred to the same protein. Alignments with 100% sequence identity were not automatically removed, because identical sequences can occur between distinct protein identifiers.

Multi-parameter strategy for edge selection

A central methodological decision was how to convert local alignment hits into comparable network edges. Because BLASTP, MMseqs2 and SWIPE differ in their search algorithms and reporting behavior, the analysis did not rely on a single raw output column alone. Instead, several alignment parameters were retained and evaluated: E-value, bit score, percent identity and alignment length.

Bit score was selected as the primary criterion for collapsing duplicate hits between the same protein pair. This decision was based on the interpretation of bit score as a normalized alignment score that is suitable for comparing alignment strength[8]. E-value was used primarily as a significance filter, because lower E-values indicate that a sequence match is less likely to occur by chance[8].

Percent identity and alignment length were retained as quality descriptors but were not used as the main edge-selection rule. Percent identity alone can be misleading because a short alignment with high identity may provide weaker evidence than a longer alignment with lower identity. Therefore, percent identity was interpreted together with alignment length, bit score and E-value.

Threshold sensitivity analysis

To avoid basing the network comparison on a single arbitrary cutoff, edge tables were generated under multiple E-value thresholds: raw unfiltered output, E-value <= 1e-3, E-value <= 1e-5 and E-value <= 1e-10. The primary downstream analysis used the E-value <= 1e-5 threshold together with highest-bit-score duplicate collapse. More permissive and more stringent thresholds were retained as sensitivity analyses to evaluate whether the observed trends depended on threshold choice.

In addition to the highest-bit-score collapse rule, a lowest-E-value version of the edge tables was generated as a robustness check. This tested whether the main network-level conclusions depended on whether edge representatives were selected by alignment strength or by statistical significance.

Construction and integration of protein similarity networks

For each tool and threshold, standardized alignment hits were converted into undirected protein similarity networks. In these networks, each node represents a protein, and each edge represents a detected sequence similarity relationship between two different proteins. This graph representation follows the general network biology framework in which biological entities are represented as nodes and their relationships as edges[3].

The per-tool edge tables were integrated by matching identical undirected protein pairs across BLASTP, MMseqs2 and SWIPE. Each integrated edge was assigned a tool-support value indicating whether it was detected by one, two or all three tools. This support-based integration was motivated by previous viral network studies, where shared genes or sequence relationships are used to reveal modular structure and robust evolutionary signal in large DNA virus datasets[2].

Network topology analysis

Network topology was analyzed for the primary E-value <= 1e-5 integrated networks. The main topology measures included node count, edge count, average degree, median degree, maximum degree, connected component count and largest component fraction. Degree was interpreted as the number of sequence similarity relationships connected to a protein, following the standard network definition of node connectivity[3].

Degree distributions were visualized using complementary cumulative degree distributions, and component size distributions were calculated to assess whether the network consisted mainly of one large connected component or multiple smaller similarity clusters. High-degree nodes were interpreted as sequence similarity hubs. They were not interpreted as physical protein–protein interaction hubs, because the network represents sequence similarity relationships rather than experimental molecular interactions.

Cytoscape visualization

Cytoscape was used for exploratory visualization of the high-confidence network structure[7]. Because the full consensus networks contained hundreds of thousands of edges, visualization was restricted to the top 10,000 highest-bit-score edges from the E-value <= 1e-5 three-tool consensus network. This subset was used only for visualization and presentation; all quantitative network statistics were calculated from the full edge tables.

Results

Alignment outputs were standardized into comparable protein-pair edge tables

The BLASTP, MMseqs2 and SWIPE all-vs-all alignment outputs were first standardized into a common tabular format. Sequence identifiers were normalized, numeric columns were parsed, and self-hits were removed before downstream network construction. After standardization, no missing numeric values and no remaining self-hits were detected in any of the three datasets. The final standardized alignment tables contained 6,550,093 BLASTP hits, 1,577,493 MMseqs2 hits and 7,710,776 cleaned SWIPE hits.

These standardized alignment tables were then converted into undirected protein-pair edge tables. In this representation, each node corresponds to a protein and each edge corresponds to a sequence similarity relationship between two different proteins. Multiple local alignment hits between the same protein pair were collapsed into a single edge. The primary collapse rule retained the hit with the highest bit score, while a lowest-E-value version was also generated as a sensitivity check.

Alignment tools differed in edge recovery and computational runtime

The three alignment tools produced substantially different numbers of protein-pair edges before thresholding. In the raw networks, SWIPE produced the largest number of unique edges, followed by BLASTP, while MMseqs2 produced the smallest edge set. However, these differences decreased under stricter E-value thresholds. At the most stringent tested threshold, E-value <= 1e-10, the three tools produced much more similar edge counts than in the raw output.

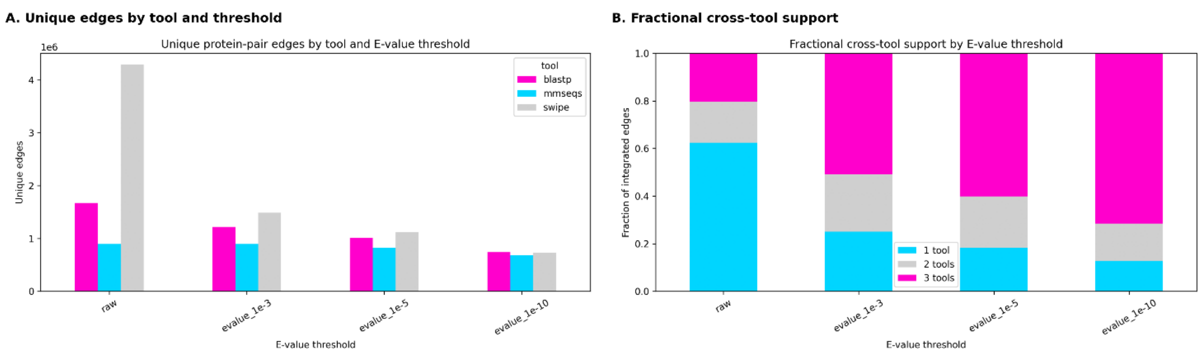

Figure 1, Tool-specific edge recovery and cross-tool support *Unique protein-pair edge counts differed between BLASTP, MMseqs2 and SWIPE, while stricter E-value filtering increased the fraction of integrated edges supported by multiple tools.*

The tools also differed strongly in runtime. MMseqs2 was the fastest method, completing the all-vs-all comparison in approximately 2 min 53 s. BLASTP required approximately 42 min 15 s, while SWIPE was substantially slower, requiring approximately 2 h 42 min. Memory usage showed the opposite trend for SWIPE: it used much less memory than BLASTP or MMseqs2. Runtime was therefore considered as an additional practical comparison, although the main downstream analysis focused on the structure and cross-tool support of the resulting protein similarity networks.

E-value filtering increased cross-tool agreement

To evaluate how consistently the three tools recovered the same protein-pair relationships, the per-tool edge tables were integrated by matching identical undirected protein pairs. Each integrated edge was classified according to whether it was supported by one, two or all three tools.

Cross-tool agreement increased strongly with stricter E-value filtering. In the raw integrated network, most edges were supported by only one tool. In contrast, after applying the primary threshold of E-value <= 1e-5, the integrated network contained 1,223,039 edges, of which 735,449 were supported by all three tools and 264,036 were supported by exactly two tools. Thus, approximately 81.7% of primary-threshold edges were supported by at least two tools. At the primary threshold, the largest support category was the three-tool consensus set. Among the two-tool categories, the BLASTP–SWIPE overlap was the largest. Among tool-specific edges, SWIPE-only edges formed the largest single-tool-specific group. This indicates that SWIPE recovered additional similarity relationships that were not reported by BLASTP or MMseqs2 under the same filtering scheme.

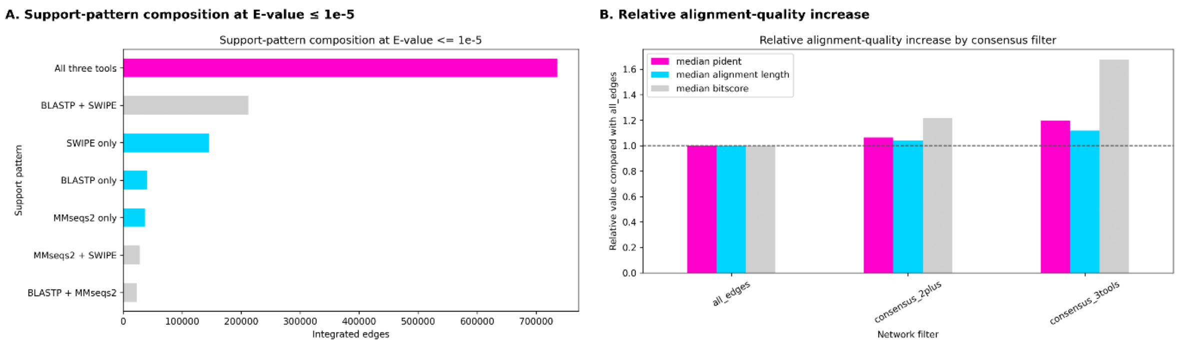

Figure 2, Primary-threshold support pattern and alignment quality *At the primary threshold of E-value <= 1e-5, most integrated edges were supported by all three tools, and consensus filtering increased median alignment-quality metrics relative to the full integrated edge set.*

Consensus-supported edges showed stronger alignment-quality metrics

Consensus filtering also affected the alignment-quality profile of the network. Compared with the full integrated edge set, requiring support from at least two tools increased the median percent identity, median alignment length and median bit score. The strict three-tool consensus network showed the strongest values for these metrics.

This suggests that multi-tool-supported edges form a more conservative and higher-confidence subset of the protein similarity network. Percent identity was not used as the primary edge-selection criterion, because high percent identity over a short alignment can be less informative than lower identity over a longer alignment. Instead, bit score was used as the primary representative alignment score, while E-value thresholds, percent identity and alignment length were retained for quality characterization.

Consensus filtering changed network topology and fragmented the largest component

The primary integrated network was analyzed at three support levels: all edges supported by at least one tool, consensus edges supported by at least two tools, and strict consensus edges supported by all three tools. Under the primary threshold of E-value <= 1e-5, the full integrated network contained 44,099 nodes and 1,223,039 edges. The two-tool consensus network contained 43,276 nodes and 999,485 edges, while the three-tool consensus network contained 42,518 nodes and 735,449 edges.

Increasing the support requirement reduced the size of the largest connected component. In the full integrated network, the largest component contained approximately 50.9% of all connected nodes. This fraction decreased to 31.0% in the two-tool consensus network and to 15.5% in the three-tool consensus network. Therefore, consensus filtering did not simply remove random edges; it changed the large-scale topology of the network by fragmenting the broad similarity graph into smaller high-confidence components.

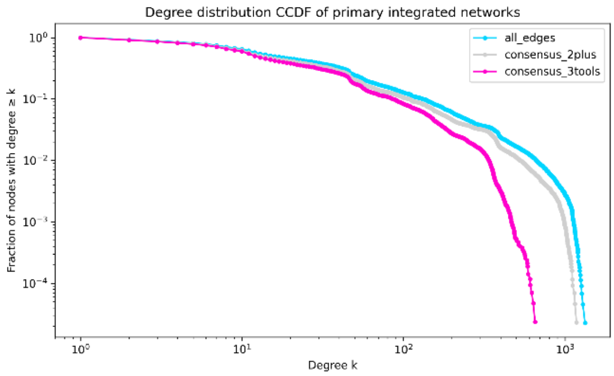

Figure 3, Degree distribution of primary integrated networks *The complementary cumulative degree distribution shows that the networks contain many low-degree proteins and fewer high-degree sequence similarity hubs, with stricter consensus filtering reducing the high-degree tail.*

The degree distribution further showed that the networks contained many low-degree proteins and a smaller number of high-degree sequence similarity hubs. These high-degree nodes should be interpreted as proteins with many sequence similarity relationships, not as physical protein–protein interaction hubs.

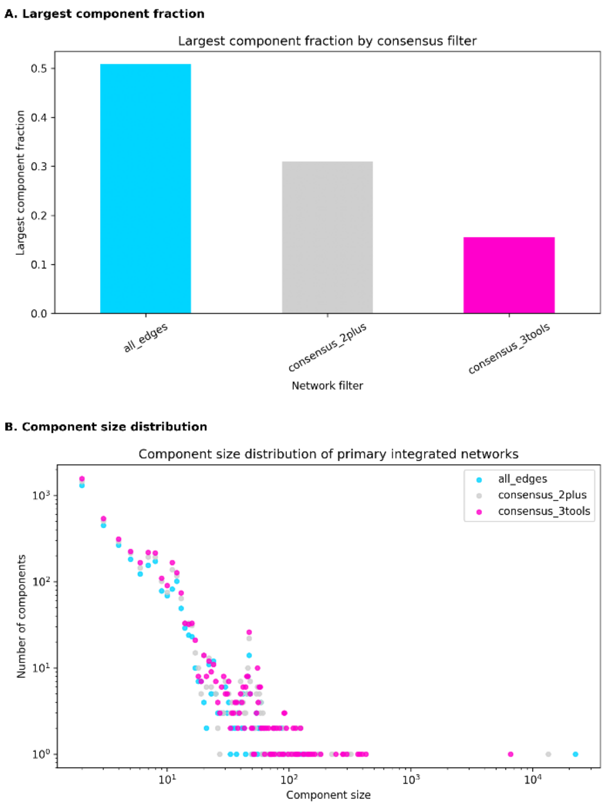

Figure 4, Network fragmentation and component structure *Consensus filtering reduced the largest connected component and revealed a component size distribution dominated by many small components and fewer large protein similarity clusters.*

The component size distribution supported the same interpretation. The high-confidence networks consisted of many small components and a smaller number of larger similarity clusters. This pattern is consistent with a protein similarity network containing multiple protein-family-like groups rather than a single homogeneous connected structure.

Cytoscape visualization confirmed the clustered structure of the high-confidence network

Cytoscape was used to visualize a high-confidence subset of the network. Because the full networks contained hundreds of thousands to millions of edges, visualization was performed on the top 10,000 highest-bit-score edges from the E-value <= 1e-5 three-tool consensus network. This subset was used only for exploratory and presentation-oriented visualization, while quantitative network statistics were calculated from the full edge tables.

The overview visualization showed that the strongest three-tool-supported edges were distributed across multiple dense and partially separated clusters. A zoomed visualization was generated for all nodes belonging to the top five degree ranks. This avoided arbitrarily selecting only five nodes when several proteins shared the same or similar high degree values. These highlighted nodes represent high-degree sequence similarity hub candidates within the exported visualization subset.

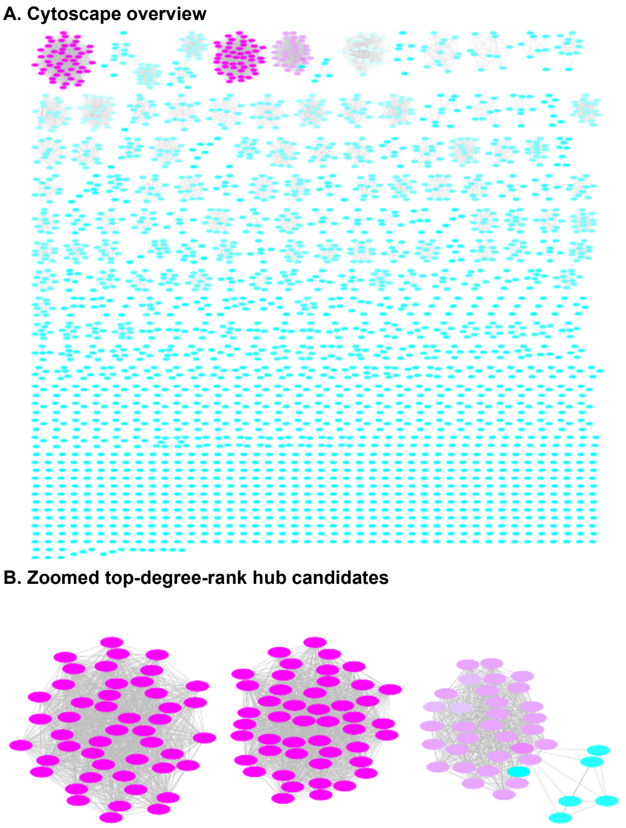

Figure 5, Cytoscape visualization of the high-confidence consensus subset *The top 10,000 highest-bit-score edges from the three-tool consensus network form multiple dense sequence similarity clusters, and the zoomed view highlights nodes belonging to the top degree ranks as sequence similarity hub candidates.*

Together, the quantitative network summaries and Cytoscape visualization indicate that the NCLDV protein similarity network contains multiple high-confidence similarity clusters. The strongest edges are enriched for cross-tool support, and stricter consensus filtering produces networks with stronger alignment-quality metrics but a more fragmented topology.

Discussion

This analysis shows that the construction of protein similarity networks is strongly influenced by the sequence comparison method and by the filtering strategy used before network construction. BLASTP, MMseqs2 and SWIPE produced substantially different raw edge sets, indicating that tool choice alone can affect the apparent density and topology of the resulting network. However, applying stricter E-value thresholds reduced these differences and increased the fraction of edges supported by multiple tools. This suggests that many tool-specific raw edges represent weaker or less consistently recovered similarity relationships, while cross-tool-supported edges form a more conservative subset.

The use of multiple alignment parameters was important for avoiding overreliance on a single score. E-value was useful for controlling statistical stringency, while bit score provided a practical criterion for selecting the strongest representative alignment between duplicate protein pairs. Percent identity and alignment length were retained as quality descriptors because percent identity alone can be misleading when alignment length differs strongly. The increase in median alignment-quality metrics after consensus filtering supports the interpretation that multi-tool-supported edges represent stronger sequence similarity relationships.

The integrated network analysis also showed that filtering does not only reduce the number of edges, but changes the topology of the network. Requiring support from at least two tools, and especially from all three tools, reduced the largest connected component and fragmented the network into smaller components. This indicates that weaker or tool-specific edges contribute substantially to connecting distant parts of the similarity graph. In contrast, strict consensus filtering highlights smaller and higher-confidence similarity clusters.

The degree distribution and Cytoscape visualization further support this interpretation. The networks contained many low-degree proteins and a smaller number of high-degree sequence similarity hub candidates. These hubs may correspond to proteins or protein families with many detectable homologous relationships within the NCLDV dataset. However, they should not be interpreted as physical protein–protein interaction hubs, because the edges in this analysis represent sequence similarity rather than molecular interaction. The Cytoscape visualization of the top 10,000 highest-bit-score three-tool consensus edges provided a useful qualitative overview of the high-confidence network structure, but quantitative conclusions were based on the full edge tables rather than the visualization subset.

Biologically, the observed clustered structure is consistent with the idea that large DNA virus protein repertoires contain both broadly shared conserved genes and more restricted protein groups. Previous studies of NCLDV and dsDNA virus evolution have emphasized that viral genomes are shaped by conserved core genes, lineage-specific genes, gene loss, gene gain and gene exchange[1,2]. A protein similarity network is therefore a useful representation because it does not require all sequences to share a universal marker gene. Instead, it can capture many local similarity relationships across a large and heterogeneous viral protein set.

At the same time, this analysis has several limitations. First, the network is based only on sequence similarity and does not directly establish orthology, function or evolutionary direction. Second, the three tools differ in their algorithms and reporting behavior, so their outputs are not perfectly equivalent even after standardization. Third, the choice of E-value threshold influences both edge count and network topology. This was addressed by testing multiple thresholds, but any selected primary threshold remains a methodological decision. Finally, the Cytoscape visualization used a high-confidence subset of the full network and should therefore be interpreted as an illustrative view rather than a complete representation of the entire dataset.

Overall, the results support the use of a multi-parameter and multi-tool strategy for constructing large viral protein similarity networks. Combining E-value thresholding, bit-score-based edge selection, alignment-quality assessment and cross-tool consensus provides a more robust workflow than relying on a single search tool or a single cutoff. This approach is especially useful for heterogeneous viral datasets such as NCLDV proteins, where similarity relationships can vary from highly conserved core proteins to weak or lineage-specific matches.

Conclusion

This project constructed and compared NCLDV protein similarity networks generated from all-vs-all BLASTP, MMseqs2 and SWIPE searches. The analysis showed that protein similarity network construction is sensitive to alignment tool choice, E-value threshold and edge-selection strategy. Raw edge sets differed strongly between tools, but stricter E-value filtering reduced these differences and increased cross-tool agreement.

At the primary threshold of E-value <= 1e-5, most integrated edges were supported by at least two tools, with the largest support category consisting of edges detected by all three methods. Consensus filtering therefore provided a more conservative high-confidence subset of the network. This filtering also improved alignment-quality metrics, but fragmented the network into smaller similarity clusters and reduced the size of the largest connected component.

Overall, the results support a multi-parameter strategy combining E-value filtering, bit-score-based edge selection, alignment-quality assessment and cross-tool consensus. This approach provides a more reliable basis for comparing large viral protein similarity networks than relying on a single tool, score or cutoff alone.

Acknowledgments

The computational results of this work have been achieved using the Life Science Compute Cluster (LiSC) of the University of Vienna.

Statement on the Use of AI Tools

The AI tools used in this assignment were GitHub Copilot in Visual Studio Code and ChatGPT. Their use was limited to supportive purposes during coding, revision, and writing.

GitHub Copilot was used mainly for code completion and for suggesting routine implementation patterns in the Python scripts. ChatGPT was used to discuss implementation ideas, review the correctness and effectiveness of proposed solutions, and improve the grammar and scientific phrasing of the report text.

The underlying solution ideas and task-specific decisions were developed independently. All AI-generated suggestions were critically reviewed before adoption. Correctness was ensured by manual code review, comparison with the assignment requirements and lecture examples, and verification against the actual program outputs. Textual suggestions were likewise revised manually to ensure accurate and appropriate scientific wording.

Code availability

The analysis scripts and report-ready summary outputs are available in the accompanying GitHub repository: *https://github.com/tiffanimre7/ncldv-protein-similarity-network*. Large raw alignment files and full edge tables are not included because of file size constraints. The repository contains the Python scripts used for alignment standardization, edge construction, network integration, summary statistics and figure generation.

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